

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
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S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 1027+400

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

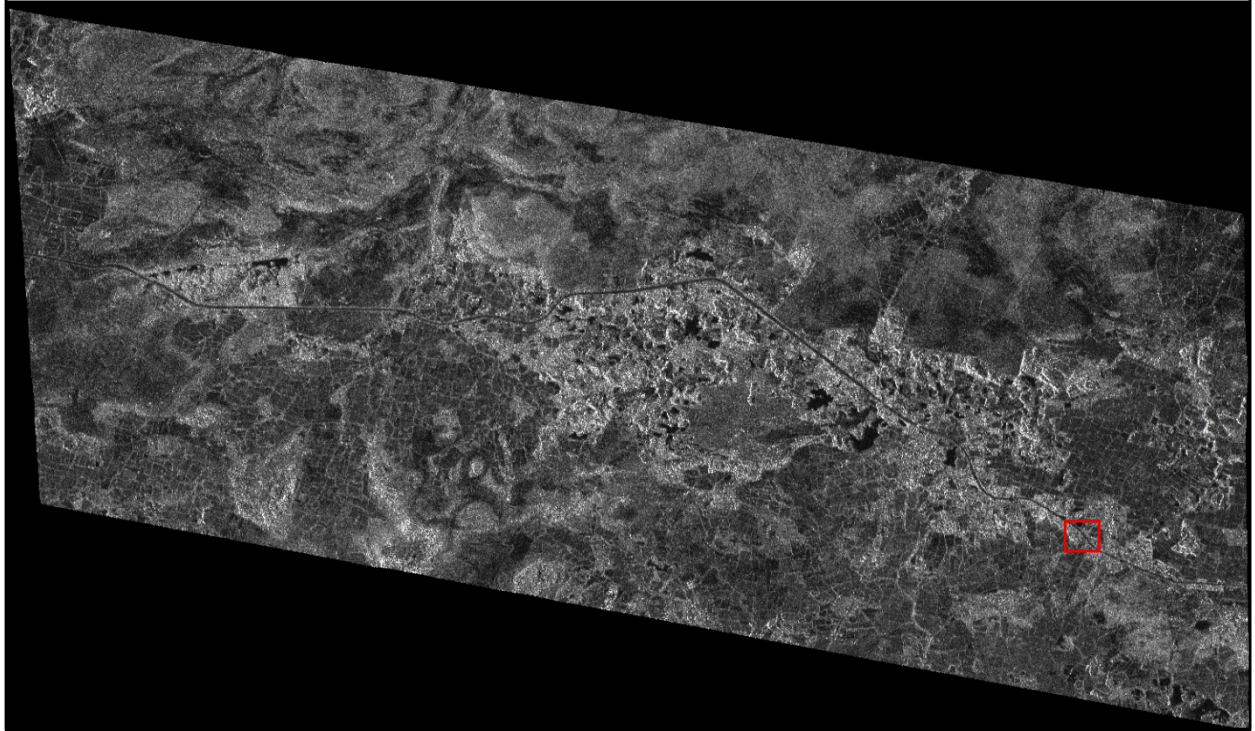
Flight Direction - DESCENDING

Polarization – VV+VH

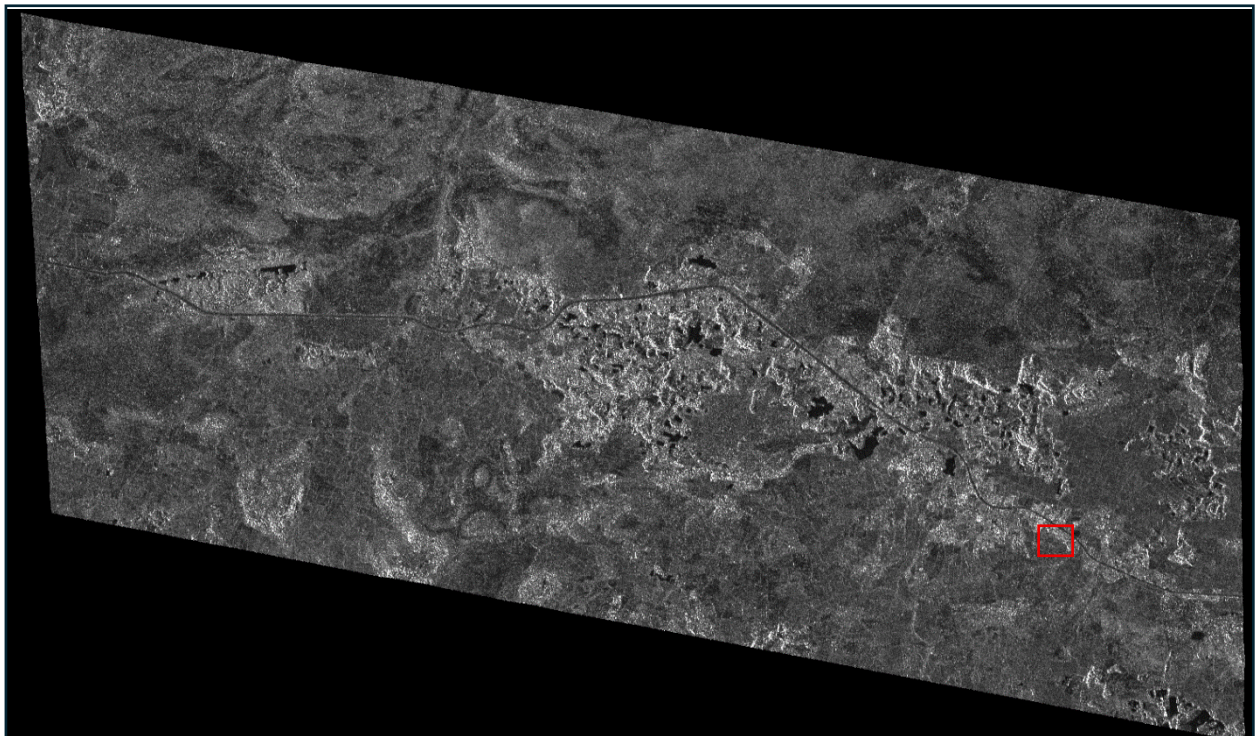
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

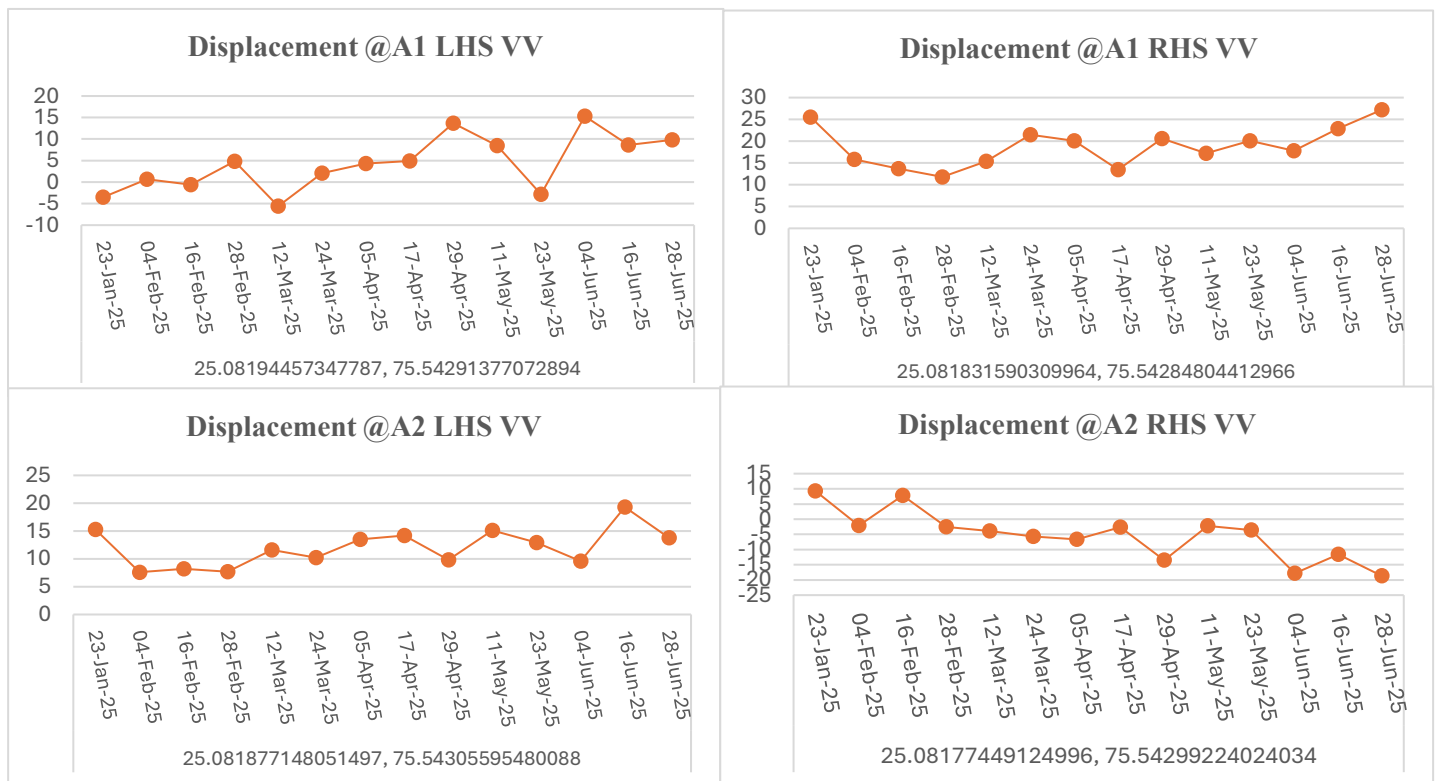
MNB – CH-1027+400

Length – 26.79+M

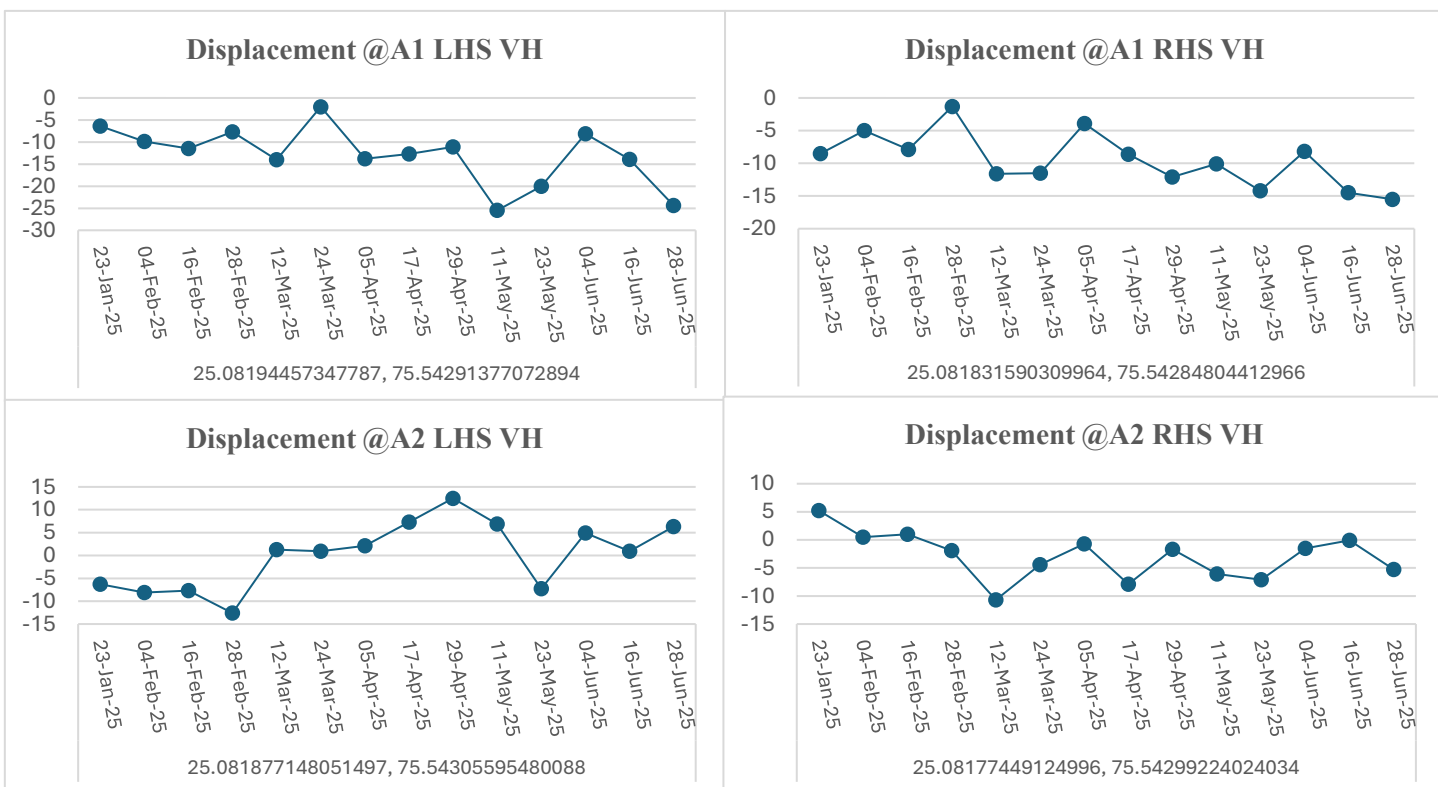


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



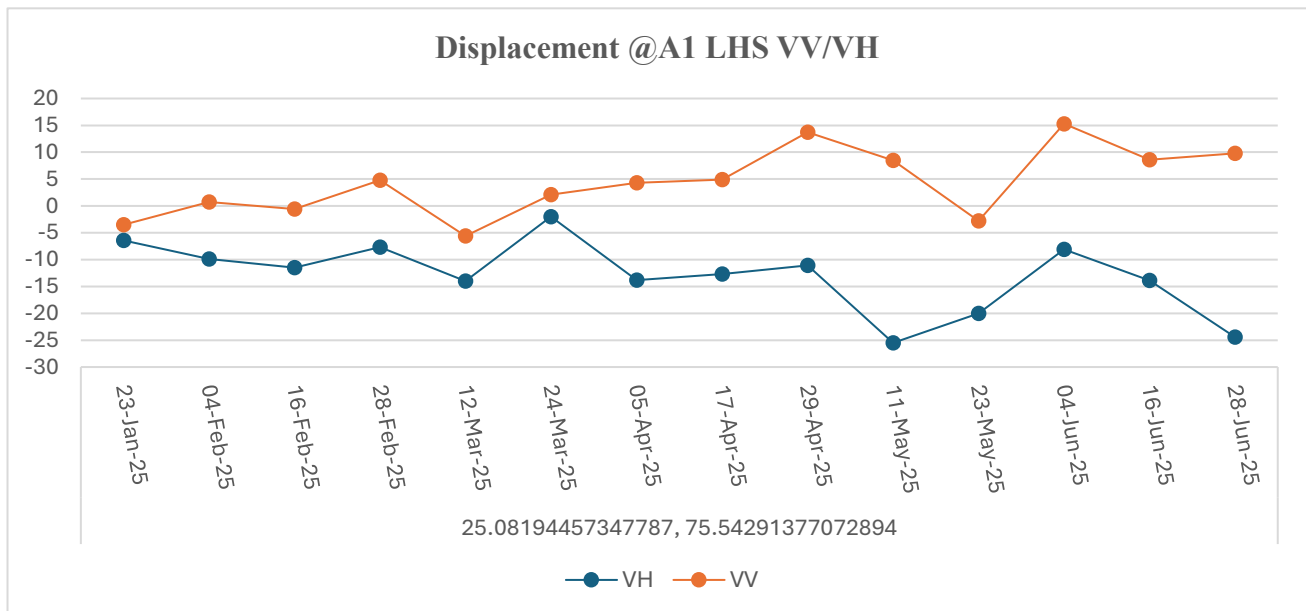
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.08194457347787, 75.54291377072894)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits alternating uplift and settlement, predominantly remaining on the positive side after mid-March. VV values range approximately from -6 mm to +15 mm.
- **Horizontal Displacement (VH)** Horizontal displacement remains consistently negative, indicating sustained lateral movement in a single dominant direction. VH values range approximately between -7 mm and -26 mm.

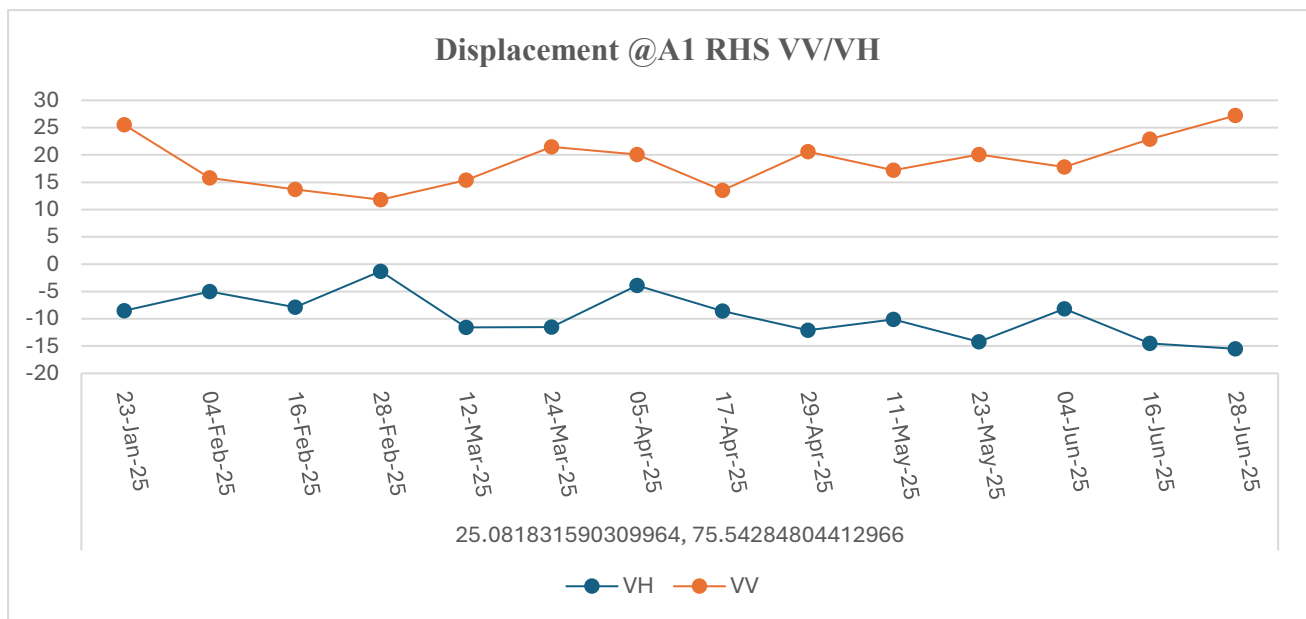
Critical Events & Anomalies:

- **March 2025:** VV briefly drops below zero while VH deepens beyond -15 mm, indicating a short phase of combined settlement and lateral shift.
- **May 2025:** VH reaches its maximum negative displacement (-26 mm), coinciding with a reduction in VV, representing the most critical horizontal movement phase.
- **June 2025:** VV peaks at approximately +15 mm, while VH partially recovers to around -8 mm, indicating temporary structural relaxation or load redistribution. VH again trends negatively toward -25 mm, while VV remains positive, suggesting persistent lateral displacement with vertical stability

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.081831590309964, 75.54284804412966)



Key Observations:

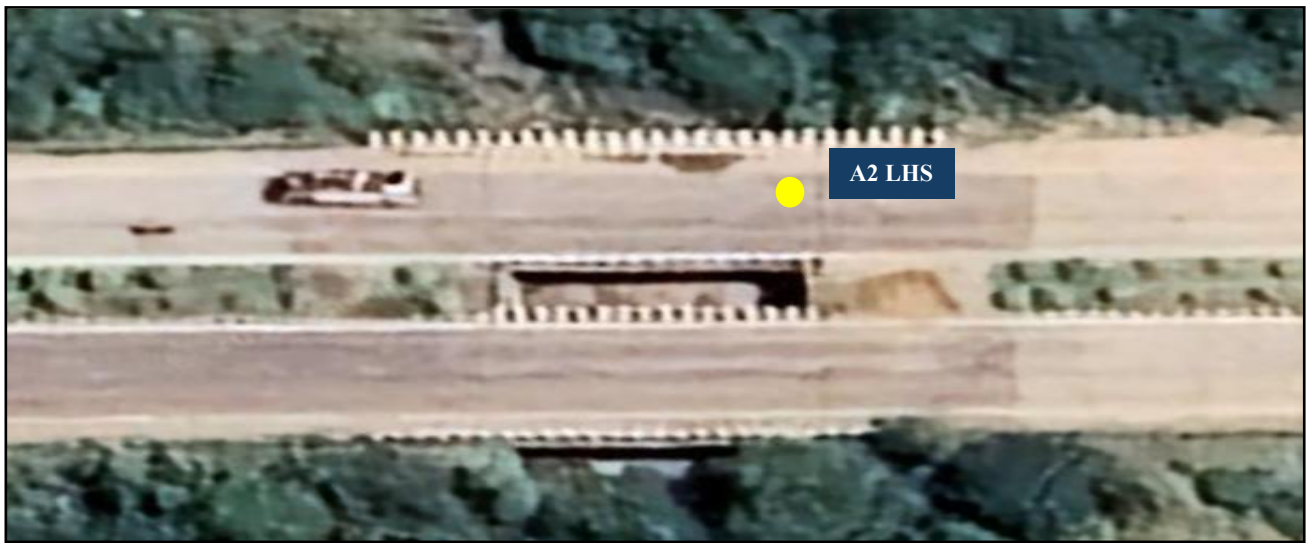
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement remains consistently negative, indicating lateral movement in a single dominant direction. Values range approximately from -4 mm to -16 mm, with moderate oscillation
- **Vertical displacement (VV)** Vertical displacement remains consistently positive throughout the monitoring period, indicating sustained uplift behaviour. VV ranges approximately from $+12$ mm to $+27$ mm, showing moderate variability but no reversal into settlement.

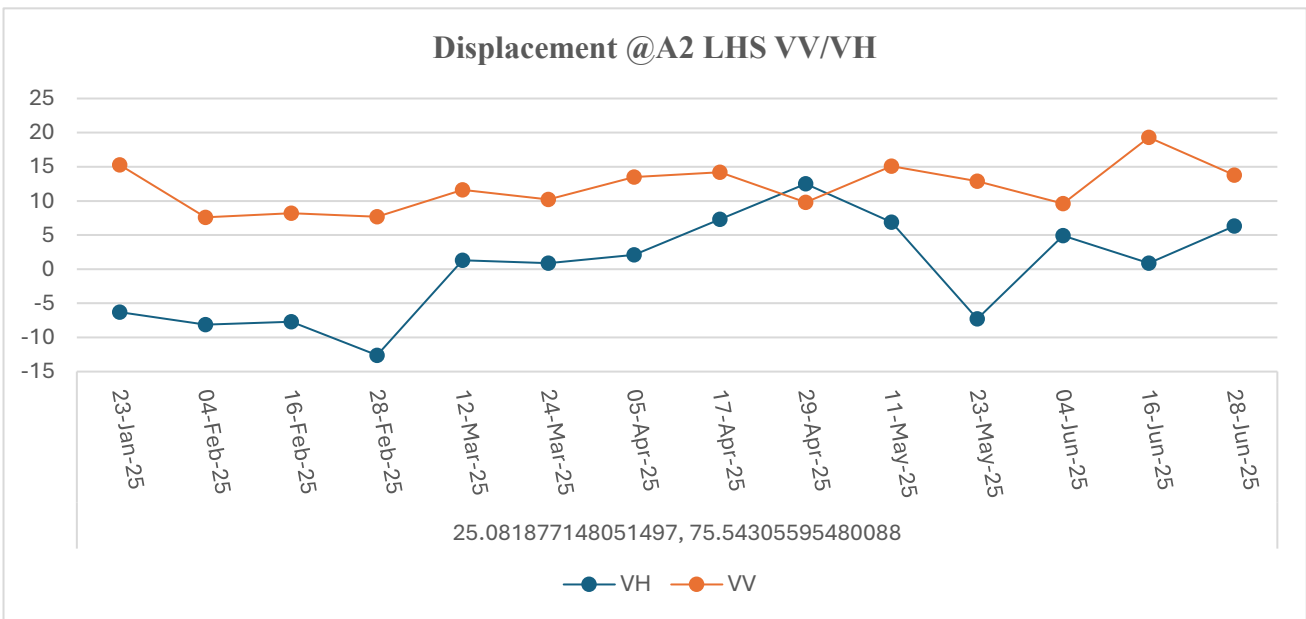
Critical Events & Anomalies:

- **March 2025:** VH reaches around -12 mm, coinciding with a mild reduction in VV — indicating coupled lateral–vertical interaction.
- **June 2025:** VV peaks near $+27$ mm, while VH remains moderately negative (-15 mm), suggesting vertical uplift dominance with restrained lateral movement.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.081877148051497, 75.54305595480088)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits a transition from negative to positive movement over the monitoring period. During January–February 2025, VH values are consistently negative, ranging approximately between –12 mm and –6 mm.
- **Vertical Displacement (VV)** Vertical displacement remains consistently positive throughout the monitoring period, indicating sustained uplift or upward movement rather than settlement.

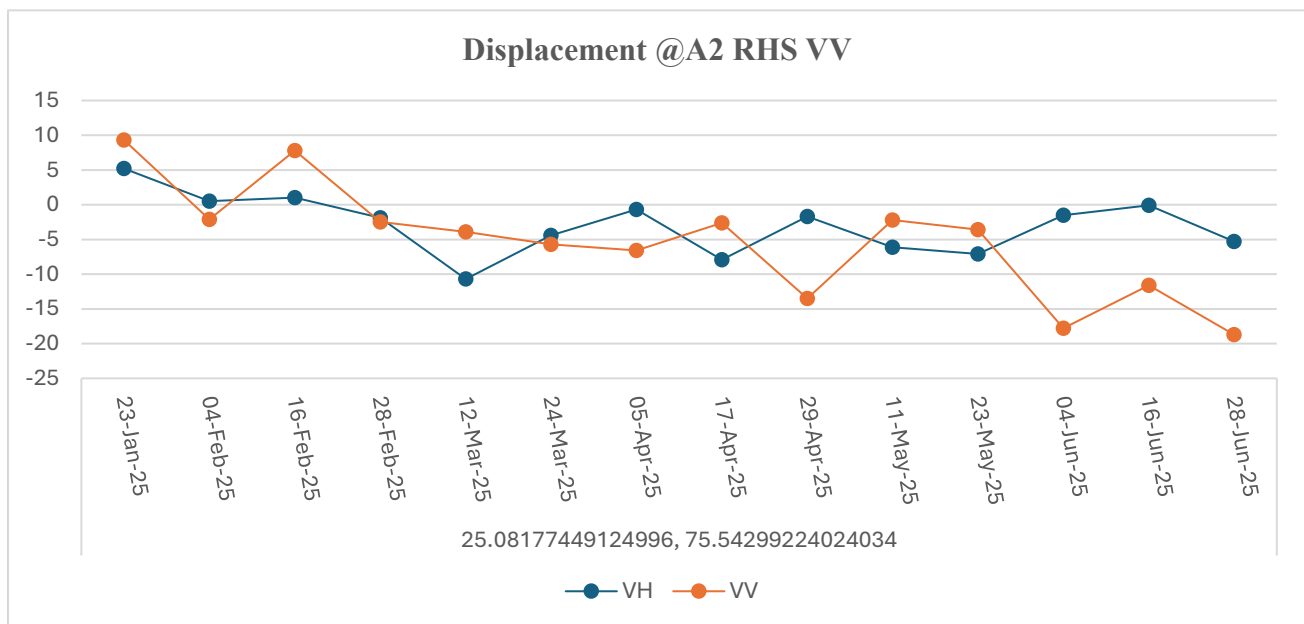
Critical Events & Anomalies:

- **February 2025:** VH reaches its maximum negative displacement (–12 mm), while VV remains relatively stable at approximately +7 to +8 mm, suggesting dominant horizontal contraction with minimal vertical variation.
- **June 2025:** VH recovers to positive values (+5 to +7 mm), while VV increases further, reaching its maximum observed uplift (+19 mm), indicating improved lateral stability alongside enhanced vertical movement.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (5.08177449124996, 75.54299224024034)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement exhibits a clear negative trend over time, indicating progressive settlement. VV values range approximately from +9 mm to -20 mm.
- **Horizontal (VH) displacement** Horizontal displacement at A2 RHS shows moderate variability around near-neutral values, with both positive and negative movements observed. VH generally ranges between approximately +5 mm and -11 mm.

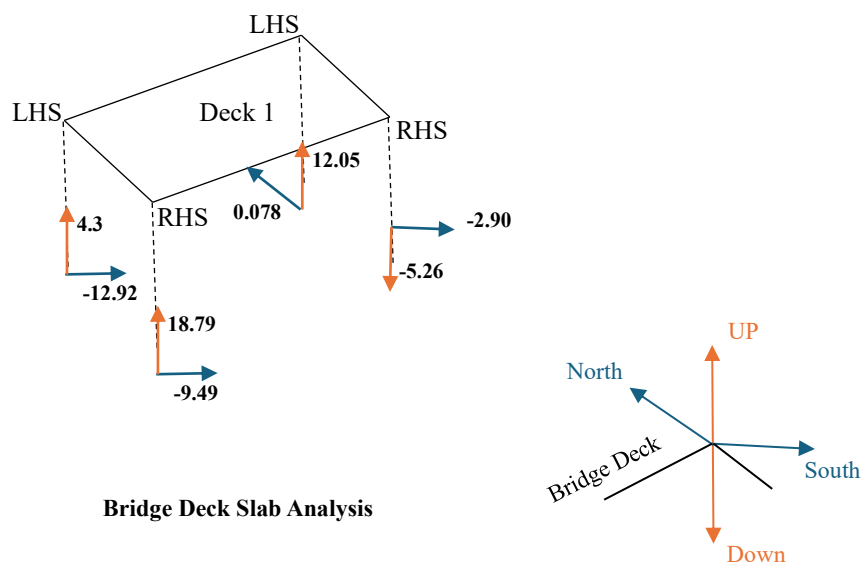
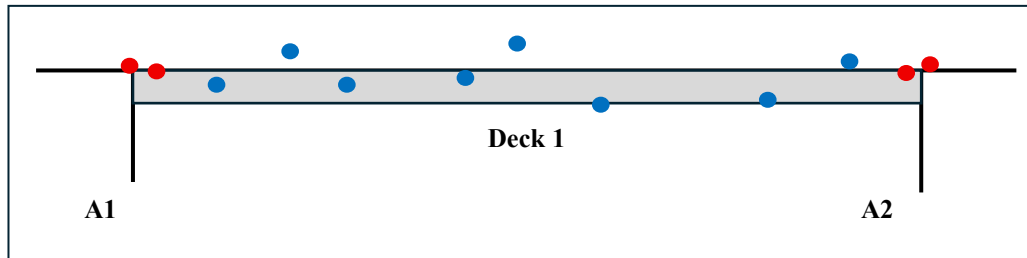
Critical Events & Anomalies:

- **February 2025:** VV initially records positive to near-neutral values (+9 mm to -2 mm), while VH remains slightly positive to near zero, suggesting relatively stable ground conditions with minimal deformation.
- **June 2025:** VV reaches its maximum negative displacement (-20 mm), representing the most critical settlement phase at this location. During this period, VH fluctuates between -7 mm and 0 mm, indicating continued lateral instability but of lesser magnitude compared to vertical movement.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits distinct non-uniform vertical displacement, indicating differential structural response across both LHS and RHS regions. The maximum uplift is observed at the central RHS deck region, reaching approximately +12.1 mm, indicating localized vertical response at the deck core.

Horizontal Displacement Behaviour:

- Horizontal displacement exhibits clear directional variation across the deck width, reflecting torsional and translational effects. The central deck region experiences minimal horizontal displacement (+0.08 mm), suggesting a relative pivot or neutral axis of rotation.

CONCLUSION:

- The bridge deck slab demonstrates a highly non-uniform deformation pattern, with significant vertical uplift and localized settlement occurring simultaneously across different regions. Overall, the observed behaviour indicates complex structural interaction involving torsion, differential support response, and asymmetric stiffness distribution, warranting continued monitoring and correlation with loading or environmental effects.